



MA 572 : Applied Multivariate Statistical Analysis

Course Code: MA 572

Course Name: Applied Multivariate Statistical Analysis

L-T-P-C: 3-0-2-4

Students intended for: B.Tech 4th year, M.Tech/MSc, Ph.D.

Elective or Core: Elective

Prerequisites: Probability and Statistics, Linear Algebra, Computer Programming

Mutual Exclusion:

Approval: 59th BoA

Course Contents

- **Recap of Prerequisites:** A quick overview of matrix computations, random variables, probability distributions & estimations. [4 Lectures]
- **Introduction to Multivariate Analysis:** Definition and scope of multivariate analysis: from univariate to multivariate thinking and the importance of data exploration; Applications in real life scenarios; Organization of multivariate data; Data cleaning and preprocessing (handling missing values, outliers, scaling, transformations); Covariance and correlation matrices (geometric interpretation); Mahalanobis distance (geometric interpretation and outlier detection); Data visualization techniques for multivariate data (scatter plots, pair plots, heatmaps, parallel coordinate plots, Andrews curves, Chernoff faces); Real-world examples. [6 Lectures]
- **Multivariate Distributions:** Multivariate normal distribution: Definition, properties (marginal and conditional distributions, linear combinations), probability density function, geometric interpretation, and applications in hypothesis testing; Wishart distribution: Definition, properties, and its role in covariance matrix estimation; Hotelling's T^2 distribution: Derivation, properties, and relation to univariate t-distribution; Multivariate t-distribution: Definition, properties, and applications in robust statistical modeling; Other multivariate distributions and their applications. [9 Lectures]
- **Multivariate Regression Models:** Introduction to multivariate regression: Multiple response variables and their modeling; Fitting and estimating multivariate regression models; Model diagnostics: Residual analysis and influence measures; Assessing model fit and significance; Generalized multivariate regression: Extending multivariate regression to handle non-linear relationships; Applications of multivariate regression in practice. [6 Lectures]
- **Canonical Correlation Analysis (CCA):** Mathematical formulation of CCA: Finding linear combinations of variables from two sets that have maximum correlation; Derivation of canonical variates and canonical correlations; interpretation of canonical correlations and canonical weights/loadings; Assumptions and limitations of CCA; Redundancy analysis; Relationship to other multivariate techniques; Applications of CCA in practice. [4 Lectures]
- **Multicollinearity and Regularization Techniques:** Detecting multicollinearity: Variance inflation factor (VIF), correlation matrix; Consequences of multicollinearity on regression models; Regularization methods: Ridge regression and Lasso regression; Tuning parameters and their impact on multicollinearity; Solving multicollinearity issues with regularization. [4 Lectures]

- **Multivariate Analysis of Variance (MANOVA) and Related Techniques:** Analysis of variance (ANOVA): One-way and two-way ANOVA; Assumptions. hypothesis testing; Multivariate analysis of variance (MANOVA): Comparing multiple groups with multiple response variables. assumptions, hypothesis testing. interpretation of results; Different tests such as likelihood ratio tests, etc.; Testing linear hypotheses in multivariate regression models. [6 Lectures]
- **Case Studies and Model Evaluation:** Case studies from different application areas; Model validation techniques: Train-test split, cross-validation, and bootstrapping ; Performance evaluation metrics: R-squared, adjusted R-squared, MSE, RMSE, precision, recall, F1-score, ROC curve, and AUC; Residual analysis, outlier detection, and influence diagnostics; Model stability, robustness, and interpretation of coefficients; Strategies for addressing overfitting and underfitting.

Lab Class:

The lab component of this course focuses on the practical application of multivariate analysis techniques through hands-on implementation with real datasets. It aims to enhance students' conceptual understanding and develop their ability to apply techniques and interpret results effectively.

Text books:

1. Johnson, R. A. and Wichern, D. W., **Applied multivariate Statistical Analysis**, PHI learning pvt ltd, 2002.
2. Anderson, T. W., **An introduction to Multi variate Statistical Analysis**, Wiley India, 2010.

References:

1. Everitt, B. S. and Dunn G., **Applied Multivariate Data Analysis**, Wiley- Blackwell, 2001.
2. Srivastava, M.S., **Methods of Multivariate Statistics**, Wiley- Interscience, 2002.
3. Tan, P. N., Steinbach M., Kumar V., **Introduction to Data Mining**, Pearson, 2014.
4. Flury, B., **A First Course in Multivariate Statistics** (Springer Texts in Statistics), Springer-Verlag, 1997.